



United International University

QUEST FOR EXCELLENCE

Department Of Computer Science & Engineering

MedGuide AI: Trustworthy AI Health Guidance and Care Navigation System for University Students

Course Title : Software Architecture

Course Code : CSE 4435

Section: (B)

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1. Introduction

MedGuide AI is a health guidance and care navigation system developed mainly for university students who often depend on online resources or AI tools when they experience common health problems. The system is designed to provide quick and understandable health-related guidance without presenting its output as a final medical diagnosis. Students can describe their symptoms through text or voice input, select symptom severity, and answer relevant follow-up questions before receiving possible condition matches. The results are presented using simple labels such as Strong Match, Partial Match, and Low Match along with urgency levels, matched symptoms, explanations, and supporting health information. This approach is intended to help users understand the possible meaning of their symptoms while also making the uncertainty of AI-generated guidance clear.

The system also provides several supporting healthcare features to make the overall process more useful and practical. Users can interact with an AI health chat, upload medical reports for simplified explanation, search for doctors, book appointments, and set medication reminders. Important guidance such as when to consult a doctor is kept visible so that users can take appropriate action when required. The main objective of **MedGuide AI** is not to replace professional healthcare services, but to work as an initial support system that can guide students towards safer and more informed decisions.

As the application deals with personal health information and health-related recommendations, its software architecture must give proper importance to system quality. Security is required to protect sensitive user data, while reliability and availability are necessary to ensure that important guidance is provided correctly and remains accessible when needed. Performance, usability, and modifiability are also important for providing quick responses, simple interaction, and easier future updates. Therefore, these quality requirements play a significant role in designing the overall architecture of **MedGuide AI**.

2. Utility Tree

Quality Attribute	Attribute Refinement	ASR Scenario	Rating (I,R)
Security	Confidentiality	During normal operation, when an unauthorized person attempts to access a student's symptoms, AI chat, medical report, or medication data, the system shall deny access and log the attempt; no unauthorized health record shall be disclosed. (H,H)	Importance : High Risk : High
	Authentication & Authorization	When a student attempts to log in during normal operation, the authentication service shall verify credentials and provide authorized access within 2 seconds ; after 5 consecutive failed attempts , further attempts shall be temporarily blocked. (H,M)	I:H R:M
	Privacy / Data Control	When an authenticated student requests deletion of stored symptoms, chats, reports, or related personal health data, the system shall remove the information from active storage and confirm the request within 24 hours. (H,H)	I:H R:H
Safety & Reliability	Emergency / Red-Flag Handling	When a student submits symptoms matching a predefined emergency/red-flag condition, the system shall prominently display appropriate urgent-care guidance before ordinary condition information; 100% of predefined red-flag test cases shall trigger the escalation mechanism. (H,H)	I:H R:H
	Uncertainty Handling	When the student's symptom information is insufficient or ambiguous, the system shall request relevant follow-up information before producing condition matches and shall not present a definite diagnosis; the behavior shall occur in 100% of predefined insufficient-information test cases. (H,H)	I:H R:H

	Graceful Degradation	If the AI analysis service becomes unavailable while a student is checking symptoms, the system shall avoid generating fabricated results, inform the student that analysis is temporarily unavailable, and continue to display safe doctor/emergency guidance within 2 seconds .(H,H)	I:H R:H
Performance	Response Time	When a student submits a completed symptom assessment during normal operating load, the health-guidance service shall return the possible-condition result page within 3 seconds for at least 95% of requests .(H,M)	I:H R:M
	Report Processing Time	When a student uploads a supported medical report of up to 5 MB during normal operation, the system shall acknowledge the upload within 2 seconds and return or begin displaying the simplified analysis within 10 seconds for 95% of requests .(M,M)	I:M R:M
Availability	Service Availability	During normal university usage, students shall be able to access symptom checking, AI health chat, doctor guidance, and reminders with at least 99.9% monthly availability .(H,M)	I:H R:M
	Disaster Recovery	When the primary application or database fails, the system shall restore the core health-guidance service within 30 minutes , with no more than 15 minutes of recoverable data loss .(H,H)	I:H R:H
Usability	Input Efficiency	When a first-time student enters symptoms using text, symptom suggestions, or voice input, at least 90% of representative users shall complete symptom entry and require follow-up questions within 2 minutes without assistance .(H,M)	I:H R:M
	Care-Guidance Understandability	After viewing an assessment result, at least 95% of representative users shall correctly identify the displayed urgency level and whether/when professional care is recommended within 10 seconds .(H,M)	I:H R:M

Modifiability	Medical Knowledge Modification	When an authorized maintainer needs to update medical guidance, urgency rules, AI prompts, or supporting-information references, the modification shall be implemented, tested, and deployed within 4 person-hours without requiring changes to the mobile/web client.(H,M)	I:H R:M
	External Service Integration	When the system needs to integrate a new doctor-booking provider or replace an existing provider that exposes a documented API, the integration shall be completed without modifying the symptom-analysis component and within 2 person-days.(M,M)	I:M R:M